Comprehensive Data Source Discovery and Integration Architecture for the 'Fessi' Hazardous Waste Knowledge Graph

Executive Summary

The digital transformation of waste management systems requires a fundamental shift from static, document-based record-keeping to dynamic, graph-based knowledge representation. The 'Fessi' knowledge graph, designed to map the complex lifecycle of hazardous materials, necessitates a robust ingestion engine capable of synthesizing heterogeneous data sources. This report delivers an exhaustive analysis of the global data landscape as it stands in early 2026, identifying the critical Application Programming Interfaces (APIs), open government datasets, and regulatory inventories required to populate this ontology.

The research identifies a bifurcated data ecosystem. The upstream domain—comprising chemical identity and physical properties—is well-served by mature, open-access APIs such as PubChem and the increasingly accessible common chemistry interfaces. In contrast, the downstream domain—waste classification, disposal regulation, and reverse logistics—remains fragmented across static PDF documents, legacy national databases, and distinct regulatory silos (e.g., RCRA in the US vs. EWC in the EU).

A pivotal finding of this analysis is the significant liberalization of dangerous goods data observed in mid-2025. The transition of the German Federal Institute for Materials Research and Testing (BAM) *Gefahrgut* database from a paid subscription model to a free, open-data service represents a strategic opportunity for the Fessi graph. This shift allows for the cost-effective, automated integration of high-fidelity transport regulations (ADR/RID/IMDG) that were previously accessible only through proprietary logistics software or manual lookup.

This report outlines a tiered data acquisition strategy. It details the technical specifications for integrating authoritative chemical APIs to establish ground truth for substance identity. It proceeds to deconstruct the complex regulatory datasets required for automated compliance, specifically analyzing the bulk XML structures of the European Chemicals Agency (ECHA) and the list-based approach of the US Environmental Protection Agency (EPA). Furthermore, it proposes a novel "bridge strategy" to link pure substance identifiers (CAS/EC numbers) to waste stream codes (EWC/LoW) by leveraging "Gold Standard" mapping tables from academic institutions and extracting unstructured disposal advice from Safety Data Sheets (SDS) via Natural Language Processing (NLP).

1. Chemical Identity and Property Data Sources

The bedrock of any hazardous material knowledge graph is the accurate resolution of chemical identities. Waste manifests, lab inventory lists, and shipping documents often employ a confusing array of trade names, archaic synonyms, and shorthand notations. The Fessi graph must act as a disambiguation engine, resolving these disparate inputs into unique, immutable chemical entities.

1.1 PubChem PUG REST API: The Central Identity Node

The PubChem database, maintained by the National Center for Biotechnology Information (NCBI), stands as the premier open-access aggregator of chemical data. For the Fessi graph, PubChem is not merely a source of information but the primary mechanism for identity resolution.

1.1.1 Architectural Capabilities and Integration Logic

The Power User Gateway (PUG) REST API provides programmatic access to over 111 million compounds, enabling the retrieval of standardized identifiers (InChIKey, SMILES), physical properties, and, crucially, Globally Harmonized System (GHS) hazard classifications.

The API functions without the requirement of an API key, reflecting its public mandate. However, this openness is balanced by strict usage policies. The system enforces a dynamic rate limit, generally capped at 5 requests per second per IP address.[1] For a high-throughput ingestion system like Fessi, which may need to process thousands of line items from a waste inventory, this necessitates an asynchronous architecture. The ingestion pipeline must implement a token bucket algorithm or a queued job system (e.g., utilizing Redis) to throttle requests and adhere to the "fair use" policy.

The URL construction for PUG REST is semantic and predictable, adhering to the pattern: https://pubchem.ncbi.nlm.nih.gov/rest/pug/<input specification>/<operation specification>/<output specification>
This structure allows for highly specific queries essential for waste characterization:

- **Synonym Resolution:** Waste producers often use colloquial names. The endpoint GET /compound/name/{name}/synonyms/JSON allows the graph to map a user input like "Muriatic Acid" to the canonical "Hydrochloric Acid" and its associated CID (Compound ID).[1]
- **Property Retrieval:** Waste treatment requires knowledge of physical states. The endpoint GET /compound/cid/{cid}/property/MolecularFormula,FlashPoint,BoilingPoint/JSON retrieves data critical for determining flammability and volatility, directly influencing storage compatibility logic in the graph.

1.1.2 GHS Classification Aggregation

One of the most powerful features of PubChem for the Fessi graph is its aggregation of GHS hazard data. Unlike a single regulatory list, PubChem aggregates classifications from multiple submitters, including chemical vendors and international agencies.

- **Endpoint:** GET /compound/cid/{cid}/classification/JSON
- **Data Insight:** This endpoint returns a hierarchical JSON object containing the "Signal Word" (e.g., Danger, Warning), "Hazard Statements" (H-Codes), and "Precautionary Statements" (P-Codes).[3]
- **Conflict Resolution:** A critical insight for the Fessi architecture is that PubChem displays *all* submitted classifications. A substance might be classified as "flammable" by one vendor and "highly flammable" by another. The graph must implement a consensus algorithm—calculating a confidence score based on the frequency of specific H-codes across submitters—to provide a reliable risk assessment to the user.

1.1.3 Systemic Limitations

While comprehensive, PubChem is an aggregator, not a regulatory authority. Data retrieved from PubChem reflects the "state of the world" as reported by contributors, not necessarily the "state of the law." Therefore, while it is excellent for physical properties and synonyms, regulatory compliance checks (e.g., "Is this a banned substance?") must be cross-referenced against authoritative lists like REACH or the EPA TSCA Inventory, rather than relying solely on PubChem's composite view.[1]

1.2 CAS Common Chemistry and Commercial Constraints

The Chemical Abstracts Service (CAS) Registry Number (CAS RN) is the lingua franca of the chemical waste industry. Virtually every environmental regulation, from the EU's Waste Framework Directive to the US RCRA, utilizes CAS numbers to define hazardous constituents.

1.2.1 The Open vs. Proprietary Divide

Accessing CAS data presents a significant challenge. The full CAS REGISTRY is a proprietary database containing hundreds of millions of substances, access to which requires expensive commercial licensing.[5] Bulk extraction or scraping of the main CAS database is strictly prohibited and technically blocked.

However, CAS maintains **Common Chemistry**, an open-access web resource and API containing approximately 500,000 substances of high regulatory and public interest.

- **Resource:** commonchemistry.cas.org
- **API Capabilities:** This API allows for the verification of CAS RNs and the retrieval of key metadata (names, InChIKeys) for the most common industrial chemicals.[6]
- **Strategic Application:** For the Fessi graph, Common Chemistry serves as a validation layer. When the system extracts a CAS number from a manifest, it can query this API to

verify that the number is active and corresponds to the expected substance name. This prevents data entry errors (e.g., a typo in a CAS number digit) from propagating through the system.

1.2.2 Handling Proprietary Data Constraints

Because the Fessi graph cannot legally replicate the CAS database structure without a license, the data model must treat the CAS RN as an *attribute* rather than a primary key. The primary key of the graph should be an open standard identifier, such as the InChIKey or the PubChem CID. The CAS RN then becomes a searchable property on the node, allowing users to query by CAS number without the system claiming to be a "CAS database replica," thus navigating the licensing constraints.[5]

1.3 GESTIS Substance Database: Occupational Safety Authority

For European operations, the **GESTIS Substance Database**, maintained by the German Institute for Occupational Safety and Health (IFA), is a gold standard for safety data. It provides curated, legally compliant information on approximately 8,800 hazardous substances, including specific occupational exposure limits and storage classes (Lagerklasse).[7]

1.3.1 Access Modalities and Terms of Use

Unlike PubChem, GESTIS does not offer a public REST API. The data is presented via a web interface.

- **Scraping Restrictions:** The terms of use explicitly prohibit the "partial or complete transfer to other information systems".[7] Automated scraping of the GESTIS website to populate the Fessi graph would be a violation of these terms and could lead to IP blocking.
- **Deep Linking Strategy:** The terms *do* explicitly permit and even encourage "linking of GESTIS substance datasheets".[7]
- **Integration Logic:** The Fessi graph should not attempt to ingest the full text of GESTIS entries. Instead, it should store the deep link to the specific substance page (e.g., https://gestis-database.dguv.de/data?name=002010 for Acetone) as a property of the chemical node.[8] This allows the Fessi application to direct users to the authoritative source for detailed medical and first-aid information, maintaining compliance while enhancing user value.

1.3.2 Data Richness

GESTIS is particularly valuable for its structured data on "Physical and chemical properties" and "Regulations." It typically reflects the harmonization of German TRGS (Technical Rules for Hazardous Substances) with EU directives. Even if the data is not ingested in bulk, manual curation or limited extraction of specific "Storage Class" codes (which are vital for

determining waste segregation compatibility) may be permissible under "citation to a limited extent" clauses, though this requires careful legal review.[8]

1.4 Comparative Analysis of Identity Sources

The following table summarizes the strategic role of each identity source within the Fessi architecture:

Feature	PubChem	CAS Common Chemistry	GESTIS	Recommended Role in Fessi
API Availability	High (Robust REST)	Medium (Limited Scope)	None (Web Interface)	**Primary vs. Secondary Source**
Identifier	CID (Compound ID)	CAS Registry Number	ZVG Number	**Identity Resolution Key**
Regulatory Status	Aggregator (Non-Authoritative)	Authority (Proprietary)	Authority (Occupational Safety)	**Compliance Validation**
Licensing	Open Data (Fair Use)	Restricted (No Bulk)	Restrictive (No Transfer)	**Data Ingestion Strategy**
Waste Relevance	High (GHS, Phys-Chem)	Medium (Verification)	High (Disposal/Storage)	**Operational Usage**

2. Regulatory Frameworks and Inventories: European Union

The European Union maintains one of the most sophisticated and rigorous chemical regulatory environments in the world. For a waste management system to function in the EU, it must align with the Classification, Labelling and Packaging (CLP) Regulation and the REACH (Registration, Evaluation, Authorisation and Restriction of Chemicals) framework. These regulations dictate not just how a chemical is sold, but how it must be classified when it

becomes waste.

2.1 ECHA C&L Inventory: The Regulatory Backbone

The Classification and Labelling (C&L) Inventory is a database maintained by the European Chemicals Agency (ECHA). It contains classification and labelling information on notified and registered substances received from manufacturers and importers. It is the definitive source for determining if a waste stream is "hazardous" under EU law.

2.1.1 Data Access and Formats

- **Absence of API:** As of the current research horizon, ECHA does not provide a dedicated public API for the C&L Inventory. While they have indicated plans to explore API capabilities in future releases of the ECHA CHEM platform, the primary method for bulk data access remains file download.[11]
- **Bulk Download Strategy:** ECHA publishes the C&L Inventory as downloadable datasets on the European Union Open Data Portal.[12] These files are typically provided in XML or CSV formats.
    - **Ingest Pipeline:** The Fessi system requires a "Batch Ingestion" pipeline. This pipeline would periodically (e.g., quarterly) check the Open Data Portal for updated dataset versions, download the massive XML dump, and parse it to update the internal graph. This "snapshot" approach is necessary due to the lack of real-time querying capabilities.[13]

2.1.2 Harmonised vs. Notified Classifications

Understanding the structure of the C&L Inventory is critical for the graph's logic. The database contains two distinct types of data:

1. **Harmonised Classifications (Annex VI):** These are classifications that have been agreed upon at the EU level and are legally binding.[14]
    - *Graph Implication:* These entries represent the "Ground Truth." If a substance has a harmonised classification, the graph must prioritize this hazard data over all other sources.
2. **Notified Classifications:** These are classifications submitted by industry (manufacturers/importers).[13] For many substances, there is no harmonised entry, and the notified entries can be highly variable.
    - *Data Variance:* It is common to find a single substance where 80% of notifiers classify it as "Eye Irrit. 2" while 20% classify it as "Not Classified" due to differences in impurities or physical form.[15]
    - *Graph Implication:* The Fessi graph cannot simply store "The Classification." It must store the *distribution* of classifications. A "Hazard Confidence Score" algorithm should be implemented. For example, if >90% of notifiers agree on a hazard, the graph treats it as definitive. If there is a 50/50 split, the system should flag the waste for manual review or default to the more conservative (hazardous) classification to

ensure safety.[15]

2.2 ECHA CHEM and REACH Registration Data

In recent years, ECHA has migrated its data dissemination to the **ECHA CHEM** platform, a redesigned interface aimed at improving the transparency and accessibility of chemical data.[11]

- **REACH Dossiers:** These dossiers contain extensive data on "Use and Exposure." They map chemicals to "Use Descriptors" such as Process Categories (PROC) and Environmental Release Categories (ERC).
- **Inferential Power:** By ingesting this data, the Fessi graph can gain predictive capabilities. If a user identifies a waste stream coming from an "Electroplating" process (PROC 7), the graph can query REACH data to identify which chemicals are registered for that use (e.g., Nickel Sulphate, Chromic Acid). This allows the system to suggest likely contaminants in the waste stream based on the industrial process, bridging the gap between upstream synthesis and downstream disposal.[13]

2.3 GHS Phrase Repositories (H and P Statements)

The output of the Fessi system often needs to be a compliant waste label. This requires the text of Hazard (H) and Precautionary (P) statements in the local language of the waste generator.

- **Data Sources:** While the codes (e.g., H200, P301) are standardized, the text strings are available in multiple languages via open-source repositories.
    - **GitHub Repositories:** Projects like mhchem/hpstatements and joyofscripting/ghs_statements provide curated JSON files mapping these codes to text in over 20 languages.[17]
    - **Official Sources:** The EU Publications Office and national institutes like Suva (Switzerland) also provide these lists in PDF or Excel formats.[19]
- **Integration:** These repositories should be loaded into the Fessi graph as static "Reference Tables." When a chemical node is linked to a hazard code (e.g., H350), the application layer can query the user's locale and retrieve the correct string from the reference table for display on the label.[17]

3. Regulatory Frameworks and Inventories: United States

For waste management operations within the United States, the regulatory landscape is dominated by the Resource Conservation and Recovery Act (RCRA) and the Toxic Substances Control Act (TSCA), managed by the EPA. The data structures here differ significantly from the EU's property-based approach, relying heavily on "listing" mechanisms.

3.1 EPA CompTox Chemicals Dashboard

The **CompTox Chemicals Dashboard** is the EPA's premier platform for chemical data, integrating diverse information on chemistry, toxicity, and exposure.

- **CPDat (Chemical and Products Database):** This dataset is uniquely valuable for "post-consumer" waste. It maps thousands of chemicals to consumer product categories (e.g., "chemicals found in floor strippers").
  - *Graph Utility:* This allows the Fessi graph to support queries based on product type. A user disposing of "expired cleaning supplies" can be guided to the likely hazardous constituents based on CPDat mappings.[21]
- **Data Access:** The EPA provides bulk data downloads (SQL dumps, CSVs) and restricted APIs. The "List of Lists" (Consolidated List of Chemicals Subject to EPCRA, CERCLA, and CAA Section 112(r)) is a critical Excel-based resource. It dictates reporting thresholds (Reportable Quantities - RQs) which trigger mandatory notifications if a specific amount of waste is released.[23]

3.2 RCRA Listed Wastes (The F, K, P, U Lists)

RCRA defines hazardous waste through a complex system of "lists" and "characteristics." This requires a sophisticated ontology in the Fessi graph.

- **P-List and U-List:** These lists cover discarded commercial chemical products. The P-list designates "acutely hazardous" wastes (e.g., Cyanides), while the U-list covers "toxic" wastes (e.g., Acetone).
  - *Mapping:* The graph must contain a direct mapping table: CAS Number <-> RCRA Code. For example, *Acrolein* (CAS 107-02-8) maps to **P003**. This is a deterministic 1-to-1 relationship for pure substances.[24]
- **F-List and K-List:** These lists cover wastes from specific and non-specific sources (e.g., "Wastewater treatment sludges from electroplating operations" - F006).
  - *Ontology Challenge:* These codes cannot be mapped directly to a single chemical. They require the graph to model "Processes" as nodes. The logic would be: Process(Electroplating) --[generates]--> Waste(F006). This requires the user to input the *source* of the waste, not just its chemical identity.[24]

3.3 State-Level Variations (California DTSC)

In the US, state regulations can supersede federal ones. California, for instance, maintains the "Cortese List" and a distinct system of waste codes that are more stringent than RCRA.

- **Data Source:** The California Department of Toxic Substances Control (DTSC) and CalEPA provide the Cortese list and active hazardous waste site lists as downloadable Excel files.[25]
- **Graph Logic:** The Fessi graph must possess "Jurisdiction" attributes. A chemical might be "Non-Hazardous" under Federal RCRA but "Hazardous" under California law. The

graph must support query parameters like ?jurisdiction=CA to return the correct State_Waste_Code alongside the Federal_Waste_Code.[25]

4. Dangerous Goods and Transportation Data

Once a waste is classified, it must be transported. This moves the domain from environmental regulation (EPA/ECHA) to transport regulation (DOT/ADR/IATA). The identifier shifts from the CAS Number to the UN Number.

4.1 BAM Dangerous Goods Database (GEFAHRGUT): The New Open Standard

A transformative development for 2025/2026 is the opening of the **BAM Dangerous Goods Database (GEFAHRGUT)**. Previously a fee-based service, the German Federal Institute for Materials Research and Testing (BAM) released this database as a free, open service in July 2025.[26]

4.1.1 Strategic Importance and Data Content

This database is authoritative for the transport of dangerous goods by road (ADR), rail (RID), sea (IMDG), and air (ICAO).

- **Data Availability:** The data is available for download in standard open formats, specifically **Excel and CSV** (tab-separated).[26]
- **Content Systems:**
  1. **UN Number System:** This provides the standardized regulatory data for all UN numbers (e.g., UN 1090 for Acetone). It includes crucial fields: Class (3 - Flammable Liquid), Packing Group (II), Tunnel Restriction Code, and Limited Quantity thresholds.
  2. **BAM Number System:** This is a more granular, internal system covering approximately 4,500 substances that are not explicitly named in the UN regulations (i.e., "n.o.s." - not otherwise specified entries). This allows for much higher precision in classifying generic chemical mixtures.[26]
- **Integration:** This dataset effectively replaces the need for scraping PDF versions of the ADR agreement or relying on expensive third-party subscriptions. It should be ingested immediately to populate the Transport_Class, Label_Requirements, and Packing_Instruction properties of the graph's nodes.

4.2 Commercial Logistics APIs

While BAM provides the *regulatory* data, commercial APIs are often needed for *operational* validation (e.g., "Will this specific carrier accept this package?").

- **UPS Dangerous Goods API:** This API allows shippers to look up hazardous material acceptance based on the specific service level (e.g., Ground vs. Air). It returns validation flags and weight limits.[28]

- **FedEx API:** Similar to UPS, this API validates dangerous goods inputs against FedEx's specific corporate policies, which may be stricter than regulatory minimums. It allows for pre-validation of shipments to prevent rejection at the terminal.[29]
- **MachShip API:** A logistics aggregator that provides a dedicated DG Lookup endpoint.
  - *Endpoint:* GET /api/faqs/dg-lookup-by-un
  - *Response:* Returns a simplified JSON object with unNumber, packingGroup, class, and subRisk.[30] This is useful for quick UI lookups (e.g., "What is UN 1230?") without querying the full internal database.

5. Waste Classification Ontologies and Mappings

The most complex task for the Fessi graph is bridging the "Chemical Identity" (CAS) to the "Waste Code" (EWC/LoW). This relationship is rarely 1-to-1; it is context-dependent.

5.1 European Waste Catalogue (EWC) / List of Waste (LoW)

The EWC is a hierarchical classification system consisting of Chapters (2-digit), Sub-chapters (4-digit), and Codes (6-digit).

- **Mirror Entries:** The core challenge is "Mirror Entries." Some wastes have two possible codes: one hazardous (marked with an asterisk) and one non-hazardous.
  - *Example:* 17 04 09* (Metal waste contaminated with dangerous substances) vs. 17 04 11 (Metal waste other than those mentioned in 17 04 09).
  - *Logic:* The distinction depends on whether the concentration of hazardous substances exceeds a specific threshold (e.g., >0.1% for carcinogens).
- **Data Sources:**
  - **EEA Datasets:** The European Environment Agency provides the official list of codes in CSV and XLS formats.[31]
  - **National Guidance:** Agencies like the UK's Environment Agency (SEPA) publish technical guidance (WM3) in Excel/PDF formats that provide the logic for these threshold calculations.[33]
- **Graph Architecture:** The graph must support **Conditional Edges**. The link between a Chemical_Mixture node and an EWC_Code node must contain a Condition property (e.g., condition: "concentration > 25%"). The query engine must evaluate this condition at runtime based on the specific mixture composition provided by the user.

5.2 University and Institutional Mapping Tables

To train the graph's inference engine, "Gold Standard" mapping tables are invaluable. Universities often publish detailed, curated lists mapping their chemical inventories to specific disposal codes.

- **ETH Zürich:** The "Sonderabfall" (Special Waste) management system is highly detailed. While they do not offer a public API, their disposal lists and "Entsorgung" databases (often available as PDFs or internal search tools) provide rigorous classifications for lab

chemicals.[35]

- **LMU Munich:** Publishes the "Entsorgung Chemikalien Liste," which directly links chemical names (e.g., Aceton) to specific disposal canisters and codes.[37]
- **University of Minnesota:** Maintains a "Chemical Waste Registry Searchable Database." This resource maps common lab chemicals to Hazard Classes (01-18) and EPA codes (D, P, U). This effectively serves as a Rosetta Stone, translating chemical names into regulatory waste categories.[39]
- **Integration Strategy:** These lists should be scraped and normalized to create a "Synonym -> Waste Code" association table. For instance, linking "Ethidium Bromide" to EWC code 16 05 06*. These empirical mappings serve as a training set for the graph, allowing it to recommend codes for similar chemicals based on structural similarity or functional use.

6. Unstructured Data Extraction: Safety Data Sheets (SDS)

A vast amount of disposal data remains locked in unstructured Safety Data Sheets (SDS), specifically in **Section 13: Disposal Considerations**. Manufacturers (Merck, Sigma-Aldrich) are required to provide this document, but the data is often in PDF format.

6.1 Parsing Strategies and Tools

Standard text mining often fails on SDSs due to their multi-column layouts and variable formatting. Specialized parsing strategies are required.

- **SDSParser:** This is a specialized Python library designed specifically for this domain. It utilizes regular expressions and layout analysis to identify section headers (e.g., "SECTION 13") and extract the contained text. It is robust enough to handle different manufacturer templates (e.g., Fisher vs. Sigma).[40]
- **pdfminer.six:** For digital-born PDFs (generated by LIMS systems), pdfminer allows for precise extraction of text coordinates, enabling the parser to distinguish between columns.
- **OCR (Tesseract):** For legacy SDSs that are scanned images, an Optical Character Recognition (OCR) pipeline using Tesseract is necessary to digitize the text before parsing.[41]

6.2 Target Data Extraction

The parsing pipeline should target specific data points within Section 13:

1. **Explicit Waste Codes:** Modern SDSs often explicitly list the EWC or RCRA code. Regex patterns (e.g., [0-9]{2} [0-9]{2} [0-9]{2}*?) can identify these reliably.[42]
2. **Disposal Methods:** NLP techniques can identify keywords like "Incineration," "Neutralization," or "Landfill."

3. **Contaminated Packaging:** The SDS often provides separate instructions for the container (e.g., "Dispose of as unused product"). This distinction is vital for accurate waste manifesting.[43]

6.3 Reliability Scoring

Data extracted from SDSs should be treated with caution. Section 13 is sometimes populated with generic boilerplate text (e.g., "Dispose of in accordance with local regulations"). The Fessi graph must implement a "Reliability Score." Data from an authoritative regulatory list (like the RCRA P-List) should have a higher score than data parsed from an SDS. The SDS data serves as a fallback or supplementary source when no regulatory listing exists.

7. Data Integration Strategy for 'Fessi'

Synthesizing these disparate sources requires a carefully designed ontology. The Fessi graph cannot be a flat database; it must be a multi-dimensional network of entities.

7.1 Entity Resolution and Linking

- **Primary Key:** The **InChIKey** (International Chemical Identifier Key) should be the immutable unique identifier for chemical nodes. It is structure-based, resolution-independent, and widely supported by PubChem and Common Chemistry.
- **Secondary Keys:** CAS RN, EC Number, UN Number. These serve as "entry points" for search but point to the InChIKey node.
- **Graph Schema:**
  - Chemical (Node) --[has_identifier]--> CAS_RN (Property)
  - Chemical (Node) --[regulated_by]--> Regulation_Entry (Node)
  - Regulation_Entry --[assigns_code]--> Waste_Code (Node)
  - Chemical (Node) --[transported_as]--> UN_Number (Node)
  - UN_Number (Node) --[requires_packing]--> Packing_Instruction (Node)

7.2 Handling Mixtures and "Mirror Entries"

The graph must handle the complexity of mixtures, which constitute the majority of hazardous waste.

- **Virtual Nodes:** The system should generate "Virtual Nodes" for specific waste streams.
  - *Example:* Mixture: Acetone_Water_50_50
  - This node inherits properties from its parents: Acetone (Flammable) and Water (Inert).
  - The graph logic calculates the resulting flash point.
  - Based on the result, it links to the appropriate EWC Code (e.g., 14 06 03* - other solvents and solvent mixtures).

8. Conclusion and Strategic Recommendations

The population of the 'Fessi' knowledge graph is a feasible but complex undertaking that requires a multi-vector data acquisition strategy. The foundational data—chemical identity—is readily available through the **PubChem PUG REST API**. The transport data layer has been significantly de-risked and democratized by the opening of the **BAM GEFAHRGUT** database in 2025.

However, the "last mile" of waste classification—linking a chemical to a specific EWC or RCRA code—remains the most challenging aspect. It requires the ingestion of bulk regulatory datasets (ECHA C&L, EPA Lists) and the intelligent parsing of unstructured documents (SDSs).

Strategic Roadmap:

1. **Immediate Action:** Implement the ingestion pipeline for the **BAM GEFAHRGUT** Excel datasets. This provides immediate, high-quality transport data (UN numbers) at zero cost.
2. **Bulk Processing:** Script the periodic retrieval and parsing of the **ECHA C&L Inventory** XML to build the European hazard classification baseline.
3. **Training Data Acquisition:** Deploy custom scrapers to harvest "Gold Standard" waste mapping tables from identified university repositories (ETH Zürich, UMN). These datasets will serve as the ground truth for training the graph's inference logic.
4. **NLP Pipeline:** Integrate **SDSParser** to process vendor SDSs, specifically targeting Section 13, to capture disposal nuances that absent from high-level regulatory lists.

By treating regulatory documents not as static text but as structured data repositories waiting to be decoded, the Fessi graph can evolve from a simple database into a predictive engine for automated hazardous waste management.

Appendix: Summary of Key Data Sources

Domain	Source Name	Access Type	Primary Data Utility	Restrictions
Identity	PubChem PUG REST	API (Open)	Synonyms, GHS, Properties	Rate Limit (5 req/s)
Identity	CAS Common Chemistry	API (Limited)	CAS Validation, InChIKey	No Bulk Scraping
Identity	GESTIS	Web (No API)	Safety Data,	No Scraping;

	Substance DB		Storage Class	Link Only
Regulation (EU)	ECHA C&L Inventory	Bulk XML/CSV	Hazardous Classification	No Real-time API
Regulation (US)	EPA List of Lists	Excel/CSV	RCRA Codes, RQs	Static Updates
Transport	BAM GEFAHRGUT	Excel/CSV	UN Numbers, ADR/RID/IMDG	**Free since July 2025**
Transport	UPS/FedEx/MachShip	API (Key)	Shipping Acceptance	Commercial Account
Waste Codes	EEA EWC List	CSV/XLS	EWC/LoW Codes	None
Unstructured	Safety Data Sheets	PDF Parsing	Disposal Advice (Sec 13)	Varies by Vendor

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